MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

Dislipidemia and hyperuricemia semiology

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Semiology: Endocrinology Module

1st Semester
2nd year Medicine
Academic year: 2026-2027

MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

DYSLIPIDEMIA

Introduction

- Any abnormality of lipoprotein metabolism leading to an increase or decrease in one or more plasma lipid fractions.
- Primary (genetic/acquired) or secondary.
- Asymptomatic+++, Symptomatic +-+
- Common conditions: 4% of the population
- Lead to complications (pancreatitis, cardiovascular risk)
- Diagnosis: clinical orientation, biological confirmation

STRUCTURE OF LIPOPROTEIN

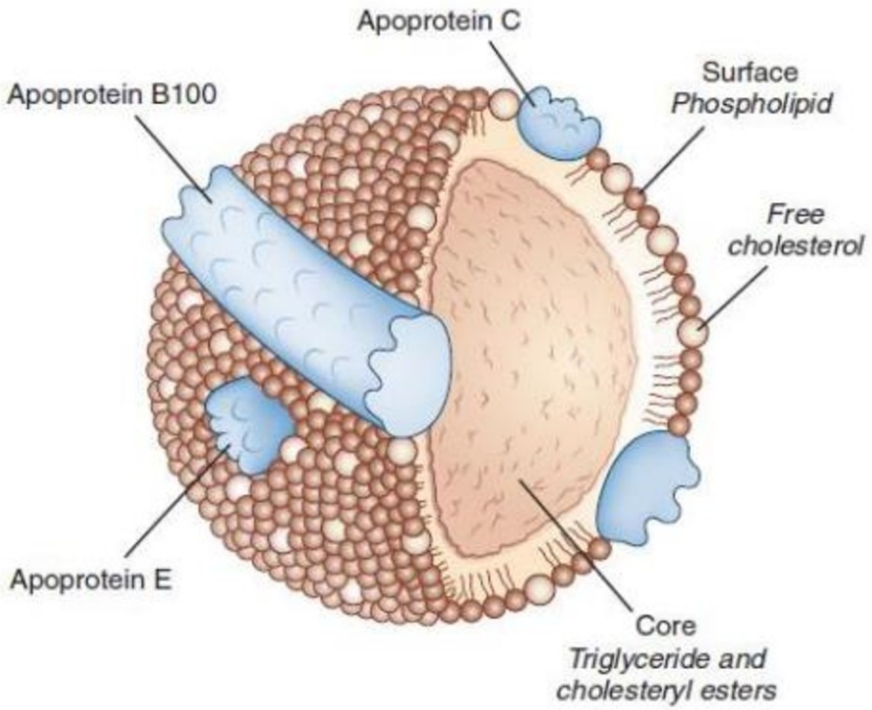
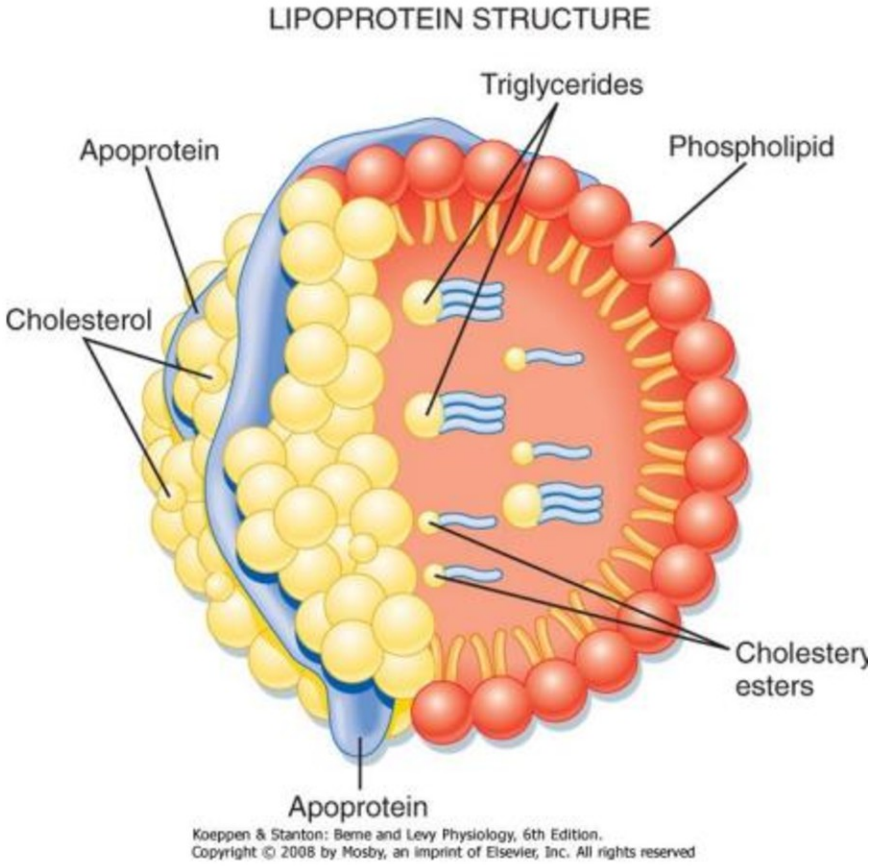


Figure 37-7 General structure of lipoproteins: schematic representation of a very-low-density lipoprotein (VLDL) particle.



Conjugated particle – Lipids: cholesterol, triglycerides
– Proteins: apoprotein (apolipoproteins)

- **Function: transport lipids and fat-soluble vitamins**

STRUCTURE OF LIPOPROTEIN

Major Classes of Plasma Lipoproteins

Type	Density (g/mL)	Origin	Major Lipids	Major Apolipoproteins
Chylomicrons	<0.95	Intestine	85% Triglyceride	B48, AI, AIV, E, CI, CII, CIII
Chylomicron remnants	<1.006	Derived from chylomicrons	60% Triglyceride 20% Cholesterol	B48, E
VLDL	<1.006	Liver	55% Triglyceride 20% Cholesterol	B100, E, CI, CII, CIII
IDL	1.006-1.019	Derived from VLDL	35% Cholesterol 25% Triglyceride	B100, E
LDL	1.019-1.063	Derived from IDL	60% Cholesterol 5% Triglyceride	B100
HDL	1.063-1.21	Liver, intestine, plasma	25% Phospholipid 20% Cholesterol 5% Triglyceride	AI, AII, CI, CII, CIII, E
Lp(a)	1.05-1.09	Liver	60% Cholesterol 5% Triglyceride	B100, apo(a)

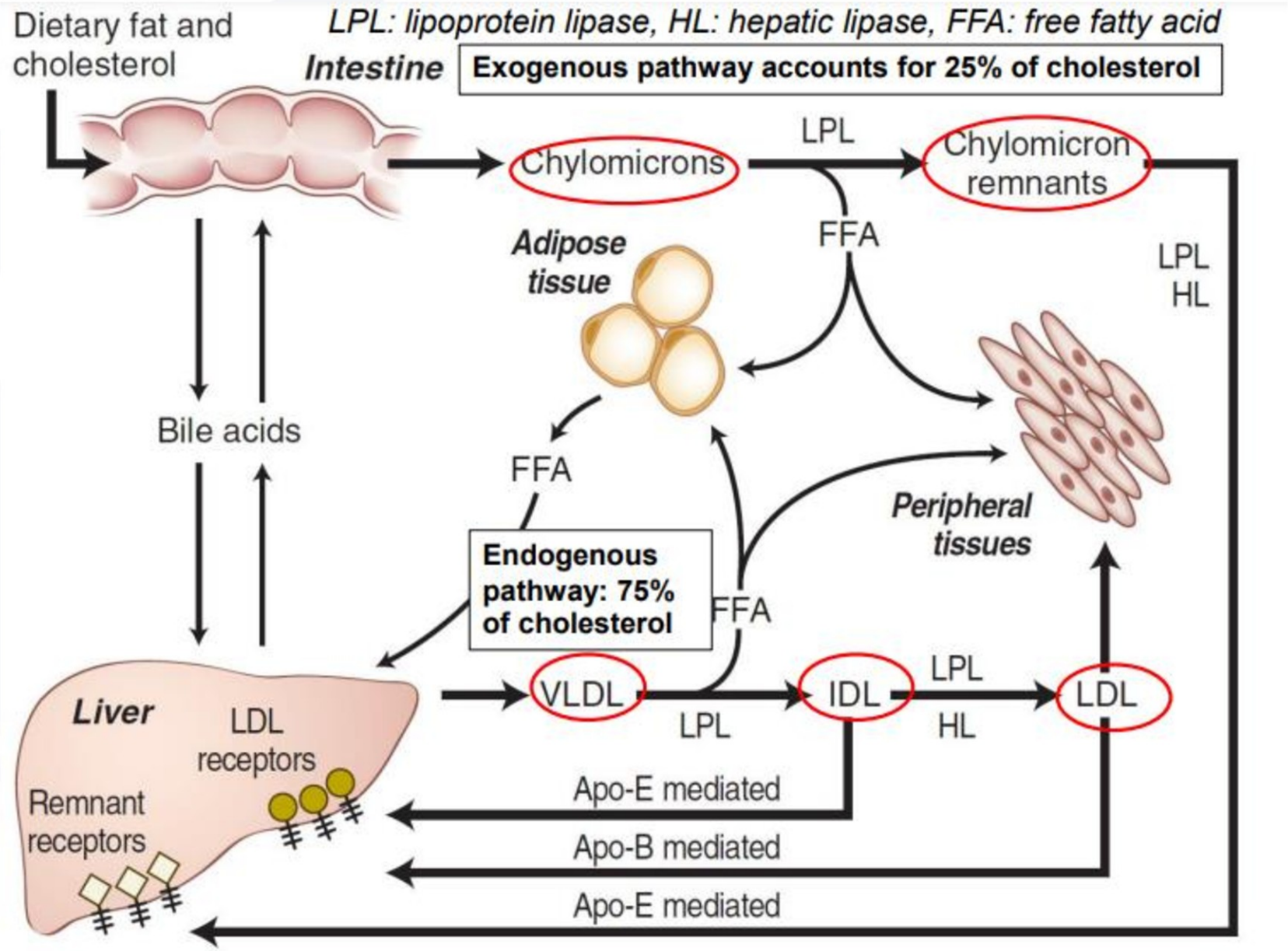
Major lipid component

- ✓ Cholesterol: LDL, IDL, HDL, Lp(a)
- ✓ Triglyceride: Chylomicron, VLDL, IDL

Major apolipoprotein

- ✓ B-100: LDL, VLDL, IDL, Lp(a)
- ✓ B-48: chylomicron
- ✓ A-I, A-II: HDL

Lipoprotein Metabolism



Fredrickson Classification of Hyperlipidemia

Type	Lipoprotein Elevated	Cholesterol	Triglyceride	Risk of Atherosclerosis
I	Chylomicrons	+	+++	Not elevated
IIa	LDL	++	Normal	High
IIb	LDL + VLDL	++	++	High
III	IDL	++	++	Moderate
IV	VLDL	+	++	Moderate
V	Chylomicrons + VLDL	+	++	Not elevated

Diagnosis of dyslipidemia

1- Clinical presentations:

- Asymptomatic
- Screening: in a subject at risk for cardio-vx or family history of dyslipidemia
- Evocative clinical manifestations
- Complications: CVx event or rarely acute pancreatitis
- Medical conditions or treatments that may lead to secondary dyslipidemia

Diagnosis of dyslipidemia

2- Clinical manifestations:

Commonly there are no signs of dyslipidemia

Diagnosis of dyslipidemia

2- Clinical manifestations:

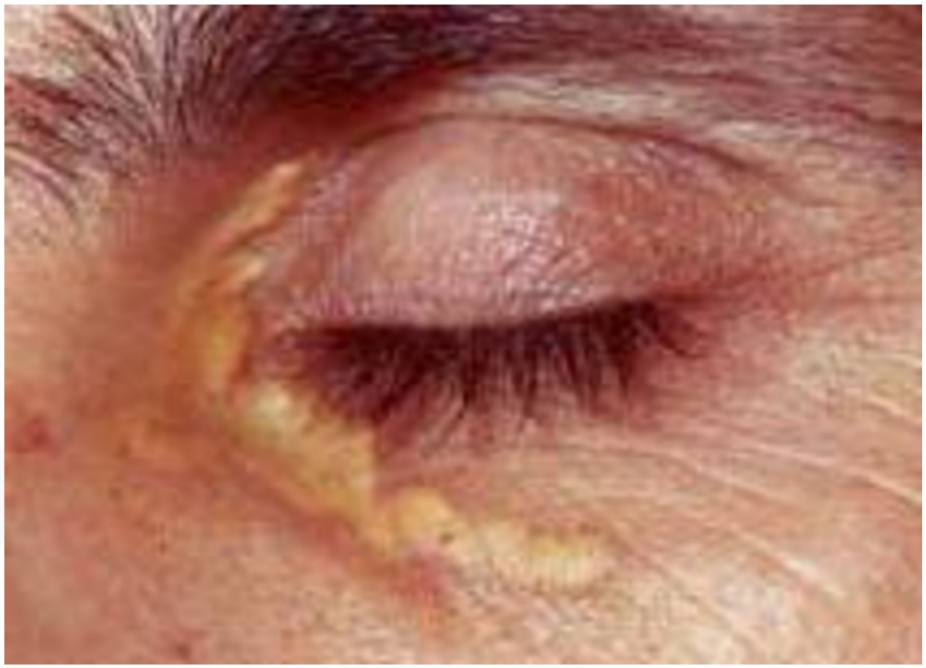
- Oohtalmologic signs :

- **Corneal arcus** occurs in patients with familial and other types of hypercholesterolemia
  In a young person, corneal arcus is very suggestive of familial hypercholesterolemia
- **Xanthelasma**
- **Lipemia retinalis**, a creamy appearance within retinal blood vessels, rarely can be seen in patients with severe hypertriglyceridemia

Diagnosis of dyslipidemia

2- Clinical manifestations:





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2- Clinical manifestations:

- Dermatologic signs :

Xanthomas

Eruptive xanthoma, which appear as small papules on buttocks and back, may occur in patients with severe hypertriglyceridemia

In familial hypercholesterolemia**, xanthomas may appear on skin covering extensor tendons of hands, Achilles tendons, and insertions of patellar tendons;**

Orange palmar creases

Diagnosis of dyslipidemia

2- Clinical manifestations:



Swiss Medical Journal

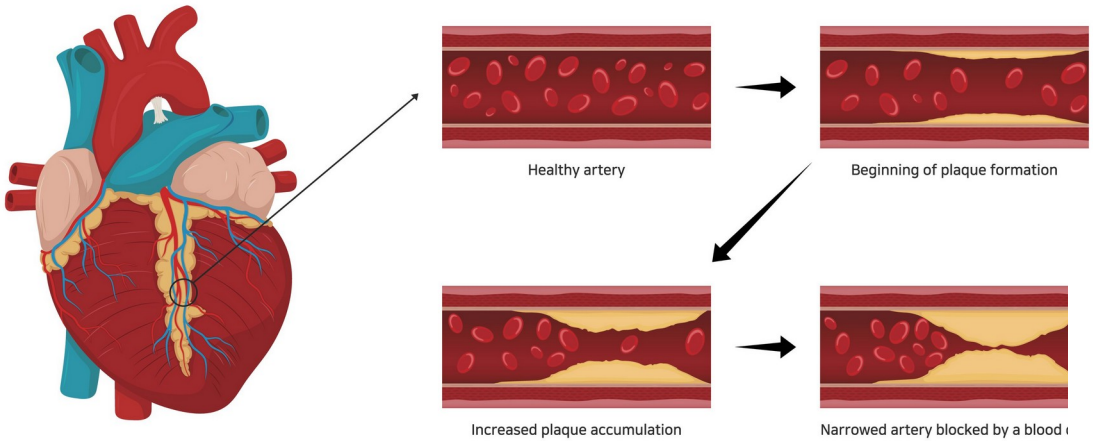
2- Clinical manifestations:

Other symptoms :

Asthenia
Post prandial sleepiness
Painful abdominal cramps that suggest pancreatitis, but may occur even in the absence of authenticatable pancreatitis.
Hepatomegaly ± Splenomegaly

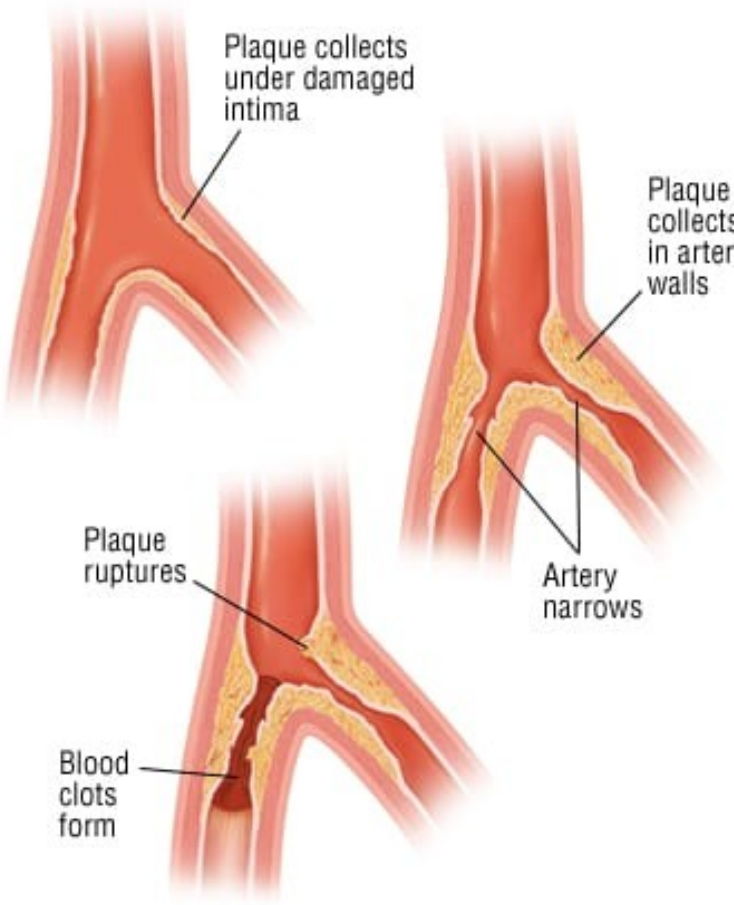
Atherosclerosis Stages

Plaque formation and growth.

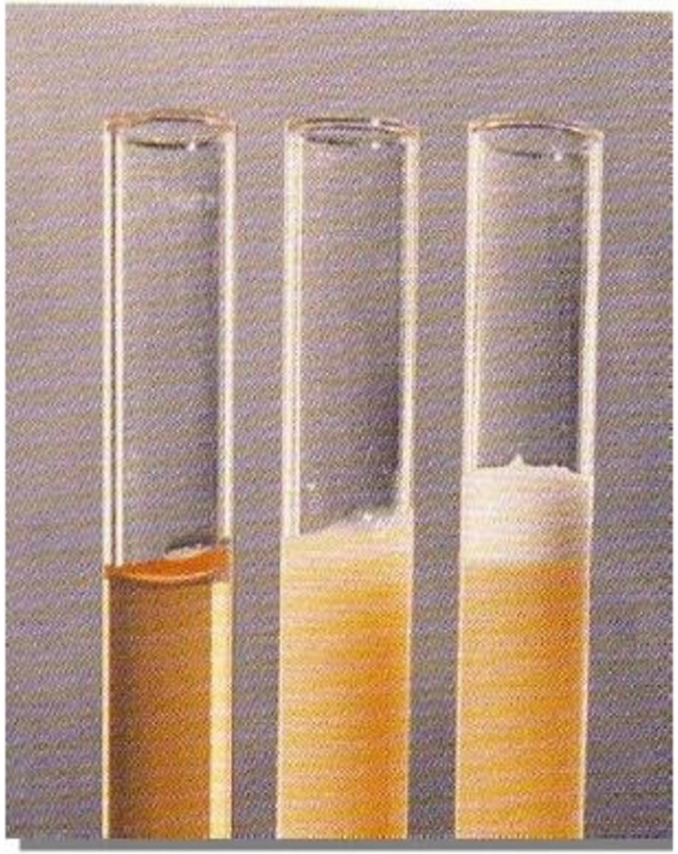


CORONARY ARTERY
CAROTID ARTERY

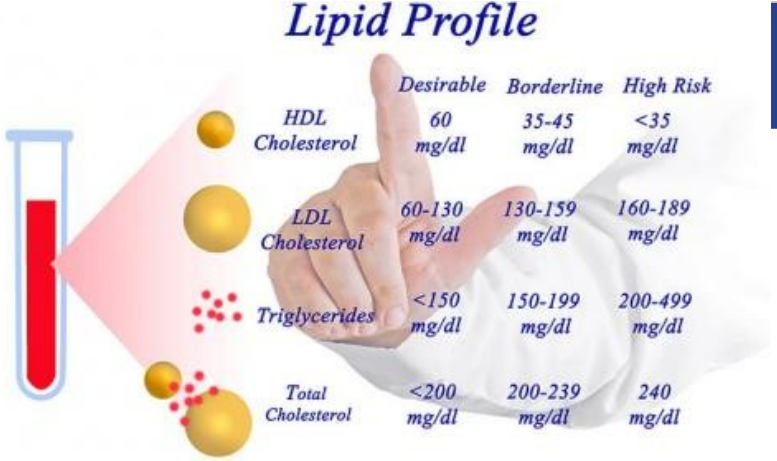
FEMURAL ARTERY



- Persistent lipemia
- After 12 hr fast
- Place in the refrigerator (4°C) for 6-10 hour
  - creamy layer: defect in *chylomicron metabolism*
  - Turbid: disorder of *VLDL catabolism*
  - Clear: disorder of *LDL catabolism*



Diagnosis of dyslipidemia



CT, TG, HDL-c determination

- LDL- is calculated according to the Friedewald formula, valid only if TG is<4g/l :

LDL= total cholesterol - HDL-c - TG/5 (in g/l)

- In a patient without Cardiovascular risk factors, the following lipid panel will be considered normal:

LDL-c < 1.60 g/l

TG < 1.50 g/l

and HDL-c > 0.40 g/l

- Recommended LDL-c values are to be defined according to the level of CV risk

A.M 76 YEARS OLD, female
NON-SMOKER
CO-MORBIDITIES: HYPERTENSION
BMI: 36 kg/m2;

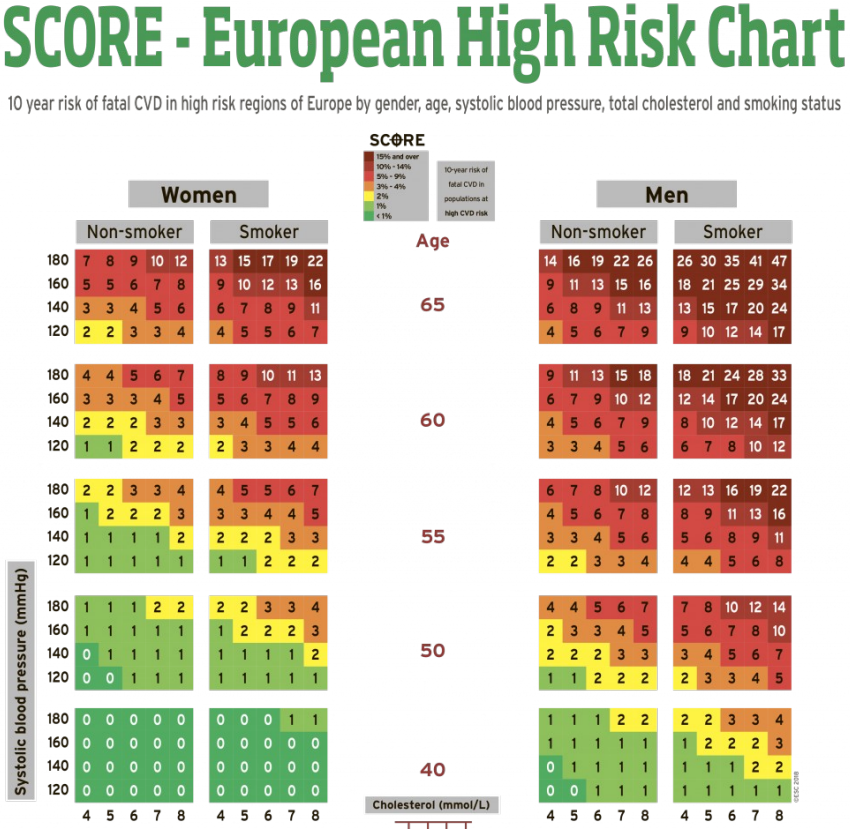
Family history of stroke (mother)

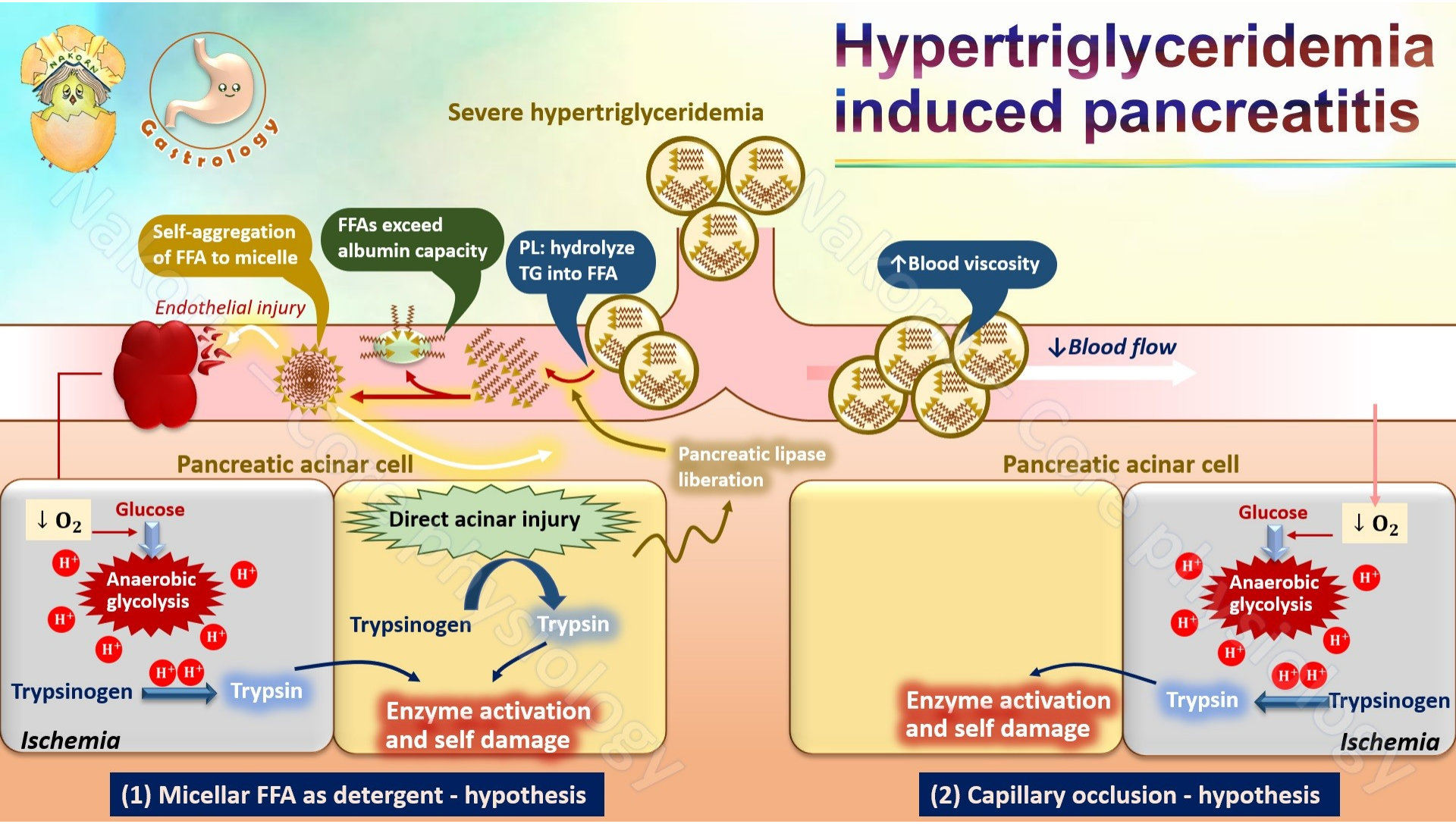
Blood test results:
A jeun glycemia=128 mg/dl (1.28 g/L)
HbA1c=6.7%
Total Cholesterol=287 mg/dl
LDLc=165 mg/dl
Apo B=140 mg/dl (<100)
Triglycerides=320 mg/dl

? Diagnosis
? Cardio-vascular risk class
? Treatment options
? prognosis

1. Diabetes mellitus– etiology?
2. Dyslipidemia type IIb
3. Obesity Stage II

Class of risk




Hypertriglyceridemia induced pancreatitis
Severe hypertriglyceridemia
Self-aggregation of FFA to micelle
FFAs exceed albumin capacity
PL: hydrolyze TG into FFA
↑Blood viscosity
Endothelial injury
↓Blood flow
Pancreatic lipase liberation
Pancreatic acinar cell
Pancreatic acinar cell
↓ O2
Glucose
Anaerobic glycolysis
Trypsinogen
Trypsin
Ischemia
Direct acinar injury
Trypsinogen
Trypsin
Enzyme activation and self damage
Enzyme activation and self damage
Glucose
↓ O2
Anaerobic glycolysis
Trypsin
Trypsinogen
Ischemia
(1) Micellar FFA as detergent - hypothesis
(2) Capillary occlusion - hypothesis

Etiologies of dyslipidemia

➢ Primary hyperlipidemias

Most commonly caused by genetic factors

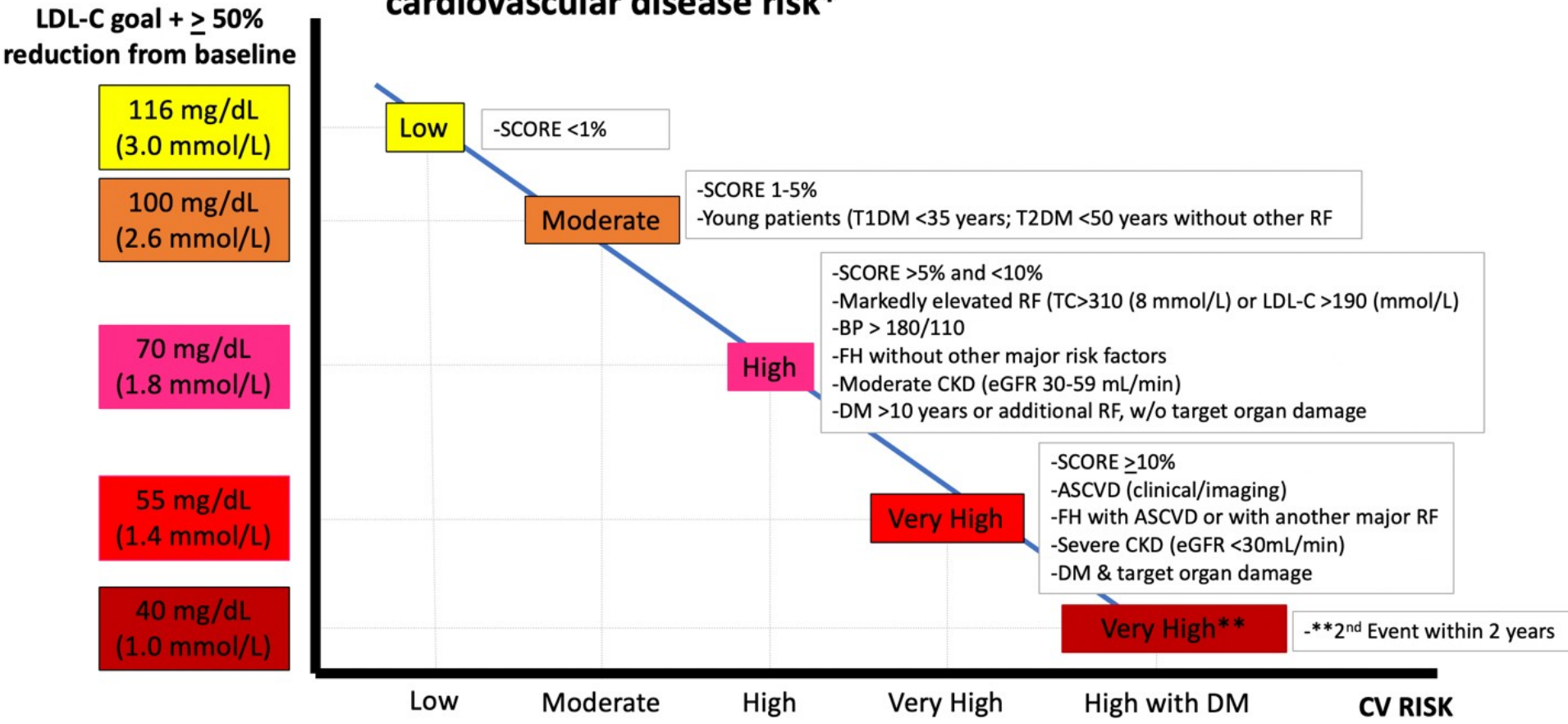
➢ Secondary hyperlipidemias:

Disorders in lipoprotein metabolism due to diseases or to iatrogenic factors.

Examples:

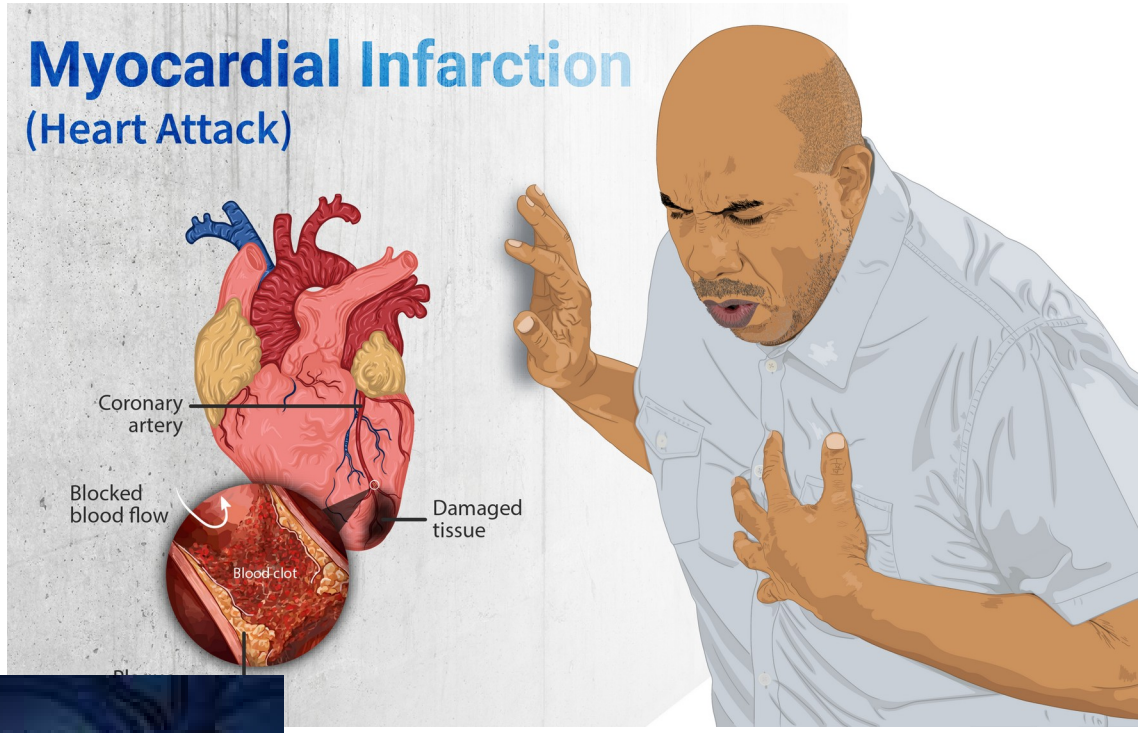
- Endocrinopathies (cushing, hypothyroidism…)
- Liver diseases
- Infectious, inflammatory, auto-immune diseases
- Iatrogenic causes: hormonal treatments, antivirals, corticosteroid+++

European Treatment goals for LDL-C across categories of total cardiovascular disease risk*



*Adapted from slideset available on www.escardio.org/guidelines which is from 2019 ESC/EAS Guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk

TREATMENT OF DYSLIPIDEMIA- PREVENTION OF CARDIO-VASCULAR EVENTS



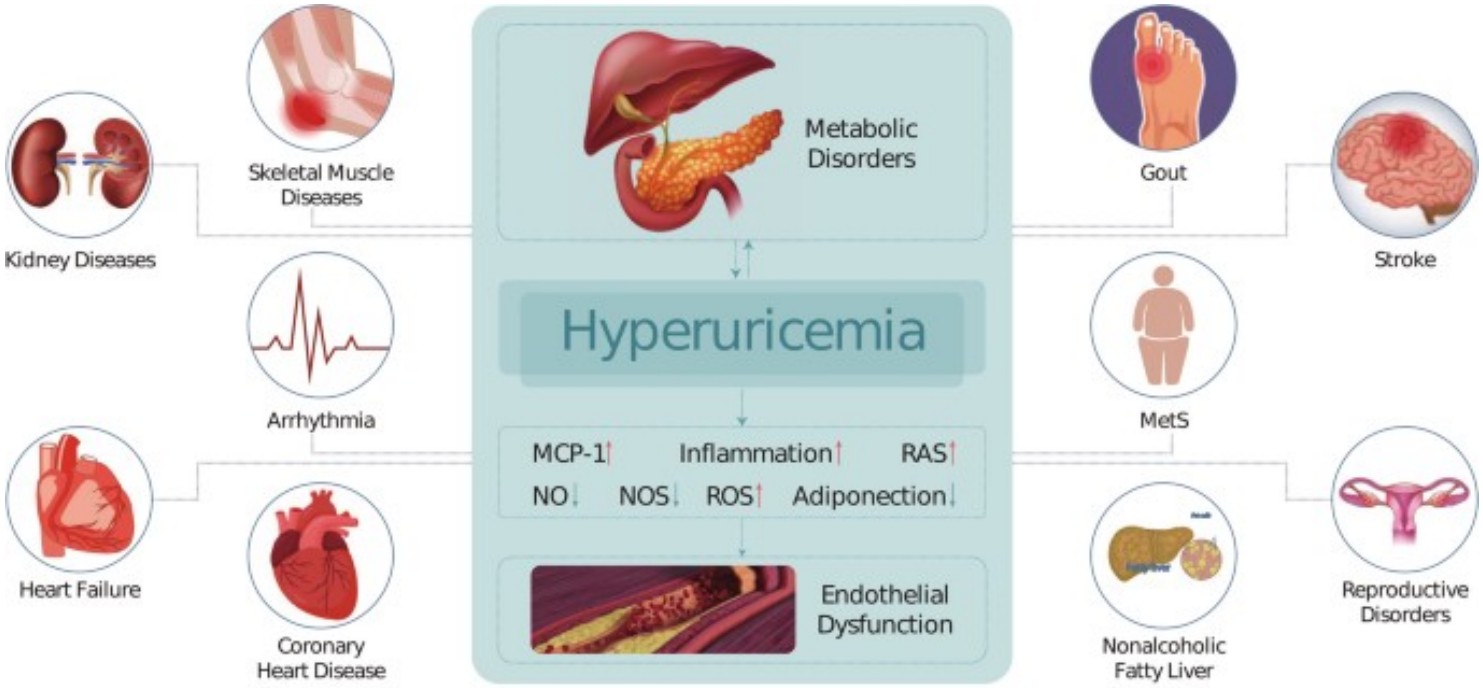


Hyperuricemia

1. Definition
2. Mechanism-pathophysiology
3. Clinical manifestation
4. Treatment

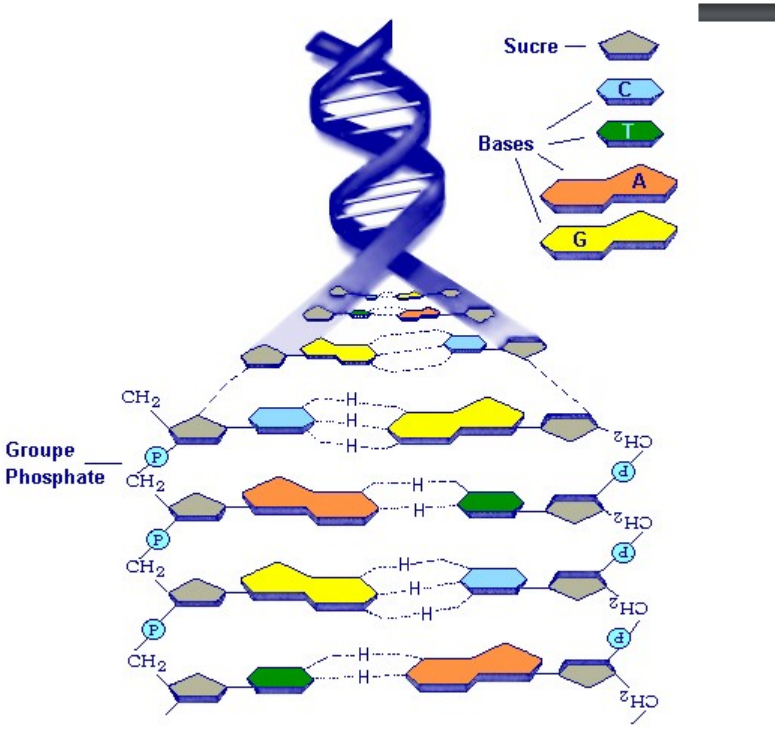
HYPERURICEMIA - DEFINITION

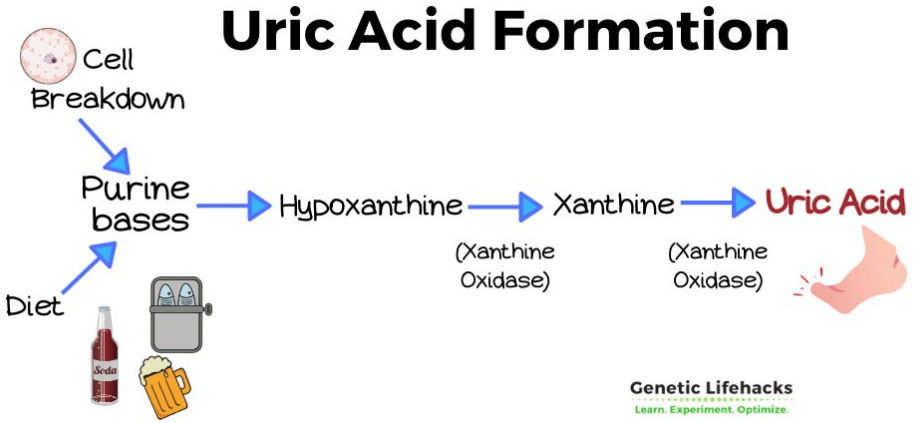
- Hyperuricemia = as an elevated serum uric acid level, usually greater than 6 mg/dL in women and 7 mg/dL in men.



The Complications of Hyperuricemia

Physiopathology





The DNA is the sourse of purines

Overproduction (10%)

Dietary purine
- Meat (beef, pork, lamb)
- Seafood (shrimps, tuna)
- Beer

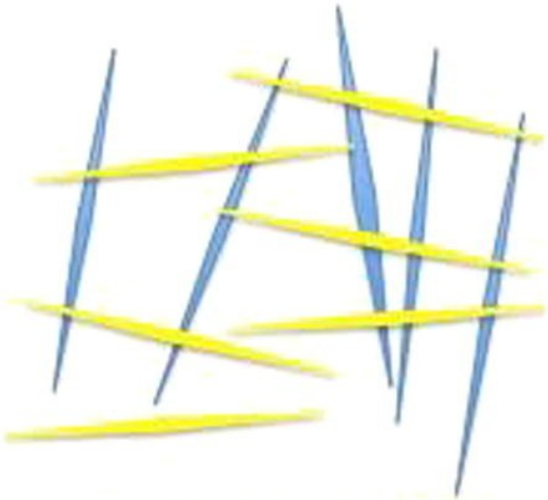
Endogenous purine synthesis
- Malignancy
- Tumor lysis syndrome

Purine salvage
- HGPRT deficiency
- PRPS deficiency

Purine breakdown
- Glycogen storage disease

Risk factors

- Male
- Age
- Obesity



URIC ACID CRYSTALS

Underexcretion (90%)

Urinary excretion
- Diuretics
- Renal failure

Urinary reabsorption
- Alcohol
- Genetic defects

Hyperuriceamia

GIT excretion

Renal excretion

CLINICAL MANIFESTATIONS OF HYPERURICEMIA

CLINICAL FORMS:
-asymptomatic hyperuricemia,
-acute gouty arthritis,
-intercritical period, and
-chronic tophaceous gout.





(a)



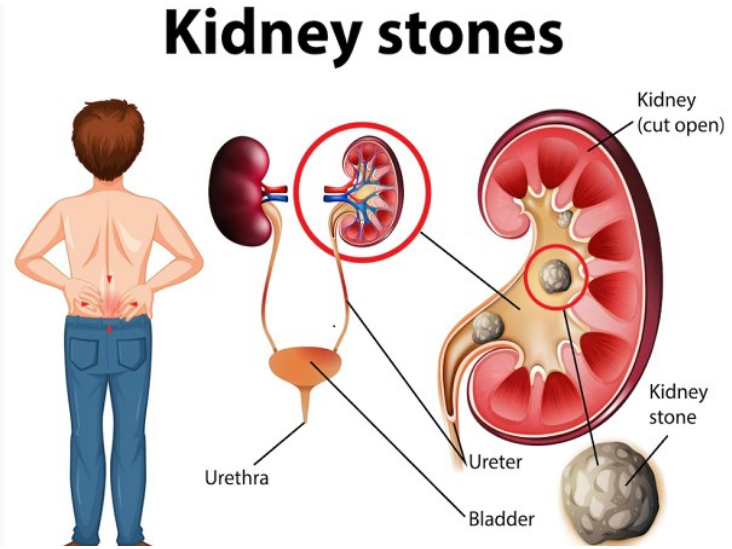
(b)

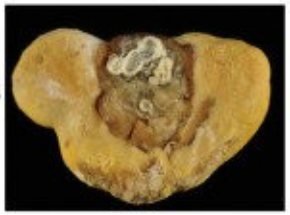
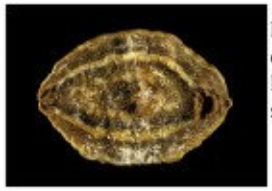


(c)

CLINICAL MANIFESTATIONS

- RENAL LITHIASIS CAN LEAD TO KIDNEY FAILURE



Morphological subtype	Stone morphology: Surface	Stone morphology: Section	Common etiology
Main component			
Uric acid **Type Ia+IIIb+IVa** (surface) **Type Ib+IVa=>IIIb** (section) **Type IVa** (nucleus=RP)			Intermittent or moderate high oxalate concentration, low diuresis Metabolic syndrome, insulin resistance, type 2 diabetes, nucleation on carbapatite RP (transcient or past hypercalciuria)
Whewellite **Type Ia+IIIb** (surface) **Type Ia+IIIb** (section) **Type Ia** (core)			Moderately high oxalate concentration, low diuresis, low urine pH, metabolic syndrome, type 2 diabetes
Weddellite **Type IIa** (surface) **Type IIIb+Ib=>Ia =>IIb** (section) **Type IIIb** (core)			Transcient hyperuricosuria and/or low urine pH + moderate hyperoxaluria + more recent hypercalciuria
Uric acid **Type IIIb** (surface) **Type IIIb/IIIa** (section) **Type Ia** (core)			Metabolic syndrome, type 2 diabetes, insulin resistance

Clinical suspicion?



Figure 1. (A) Tophaceous gout with multiple tophi on both hands. (B) The left hand X-ray shows punched-out erosions (thick arrows) with overhanging edges (arrowhead) and soft tissue nodules (thin arrows), findings suggestive of gout. Reproduced from Barajas-Ochoa et al